ACE
Engineering College

An Autonomous Institution

Ankushapur (V), Ghatkesar (M), Medchal Dist. – 501 301

Assignment 2(A): Utilities, Collections, and Concurrency

Last Date for Submission: 06-04-2026 **Course**: IT405PC Java Programming

Total Marks: 2.5

Unit III & Unit IV (Utility Classes, Collections, and Multithreading)

1. Briefly explain the use of **StringTokenizer** and **Scanner**. Write a program that uses the **Calendar** class to display the current time and the **Random** class to generate five unique lottery numbers between 1 and 100.

2. Compare **ArrayList** and **LinkedList**. Write a program to store a list of product names in an **ArrayList**, sort them alphabetically using **Collections.sort()**, and print them using an Iterator.

3. What is the difference between **Comparable** and **Comparator**? Create a class **Employee** (id, name, salary) and write a program to sort a list of employees based on their salary using a custom **Comparator.**

4. Explain the Thread Life Cycle. Write a program to create two threads: one that prints even numbers and another that prints odd numbers up to 20, using the **Runnable** interface.

Note:

- All assignments must be submitted on or before the mentioned date; late submissions will not be accepted, and students who fail to submit will be awarded 0 marks.

Presentation Guidelines:

- Use A4 sheets. Write your Name, Roll Number, Section, and Department clearly on the first page.
- Maintain logical flow and neat presentation throughout.

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Assignment 2(B): Database Connectivity and GUI Development

Last Date for Submission: 06-04-2026 **Course**: IT405PC Java Programming

Total Marks: 2.5

Unit IV & Unit V (JDBC, Swing, and Event Handling)

1. Describe the JDBC Architecture and the four types of drivers. Write the standard steps (and code snippets) required to establish a connection to a database and execute an **INSERT** query.

2. Explain the MVC (Model-View-Controller) architecture in the context of Swing. Create a simple GUI window using **JFrame** that contains a **JLabel** and a **JButton**.

3. Compare **FlowLayout**, **BorderLayout**, and **GridLayout**. Write a program that arranges five buttons in a **BorderLayout** (North, South, East, West, Center) to demonstrate their positioning.

4. Explain the Delegation Event Model. Write a Swing application where clicking a button changes the background color of the frame, demonstrating the use of an **ActionListener** or an Adapter class.

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